

IILM University, Greater Noida.

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YOUR PARTNER IN CAREER GROWTH

Career Link

An AI-Powered Off-Campus Job Search & Hiring Platform

INTRODUCTION

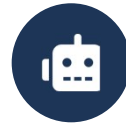
What is Career Link?

A centralized platform that connects students seeking off-campus opportunities with companies looking for skilled, job-ready candidates.



Curated Company Listings

Detailed company profiles, job descriptions & interview insights



AI-Powered Recommendations

Matches resumes to relevant job openings based on skills



Subscription & Alerts

Real-time updates on new job openings and interview insights



Smart Hiring for Companies

Profile creation, candidate tracking & AI-driven matching

THE CHALLENGE

Problem Statement



Off-campus hiring today is fragmented, slow, and hard to navigate.



No Centralized Platform

Students lack one reliable place to find off-campus openings



Limited Reach for Companies

Recruiters struggle to access skilled talent beyond campus drives



Job-Role Mismatch

Poor alignment between candidate skills and job requirements



No Interview Insight

Little visibility into a company's hiring/interview process

GOALS

Project Objectives



Bridge Students & Companies

A structured platform for discovery on both sides



Enhance Search Efficiency

AI-powered job recommendations based on resumes



Deliver Meaningful Insights

Company profiles, job details & interview data



Streamline Hiring

AI-driven candidate matching for recruiters



Keep Users Updated

Subscriptions for real-time job alerts & trends



Fix Job-Role Mismatch

Address common job market pain points



Modern, Easy-to-Use UI

Smooth navigation and accessibility for all users

BOUNDARIES

Scope of the Project



– Role-based registration & login for students and companies



– Company listing with search & filter functionality



– Interview / job posting and browsing



– Subscription system for real-time updates



– Review and rating system for companies



– Scalable architecture for future add-ons



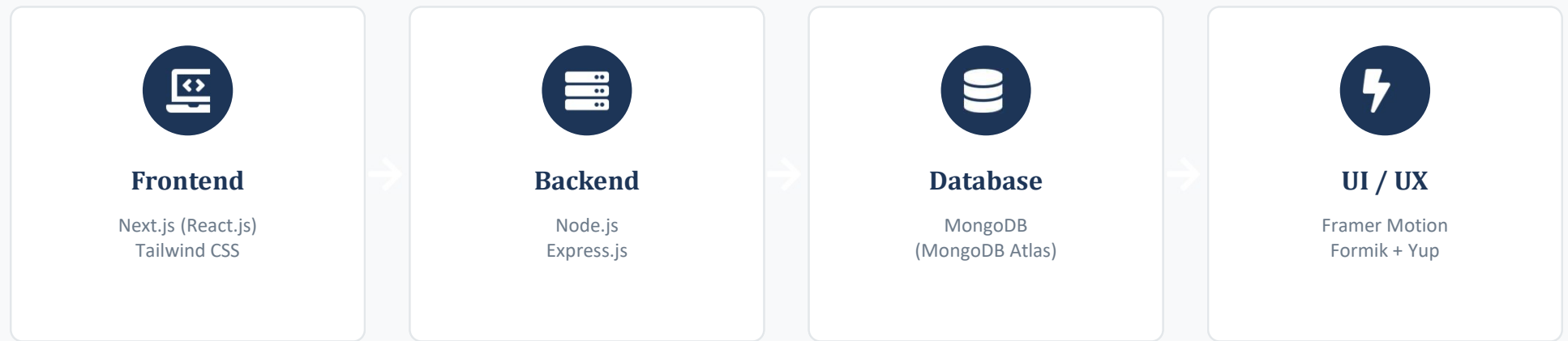
Built to Grow

Career Link is architected with a modular, scalable foundation. Planned future capability includes:

- Admin analytics dashboard
- Deeper AI-driven matching
- Third-party API integrations
- Expanded notification system

MERN STACK

Technology Stack



Supporting Tools

VS Code

GitHub

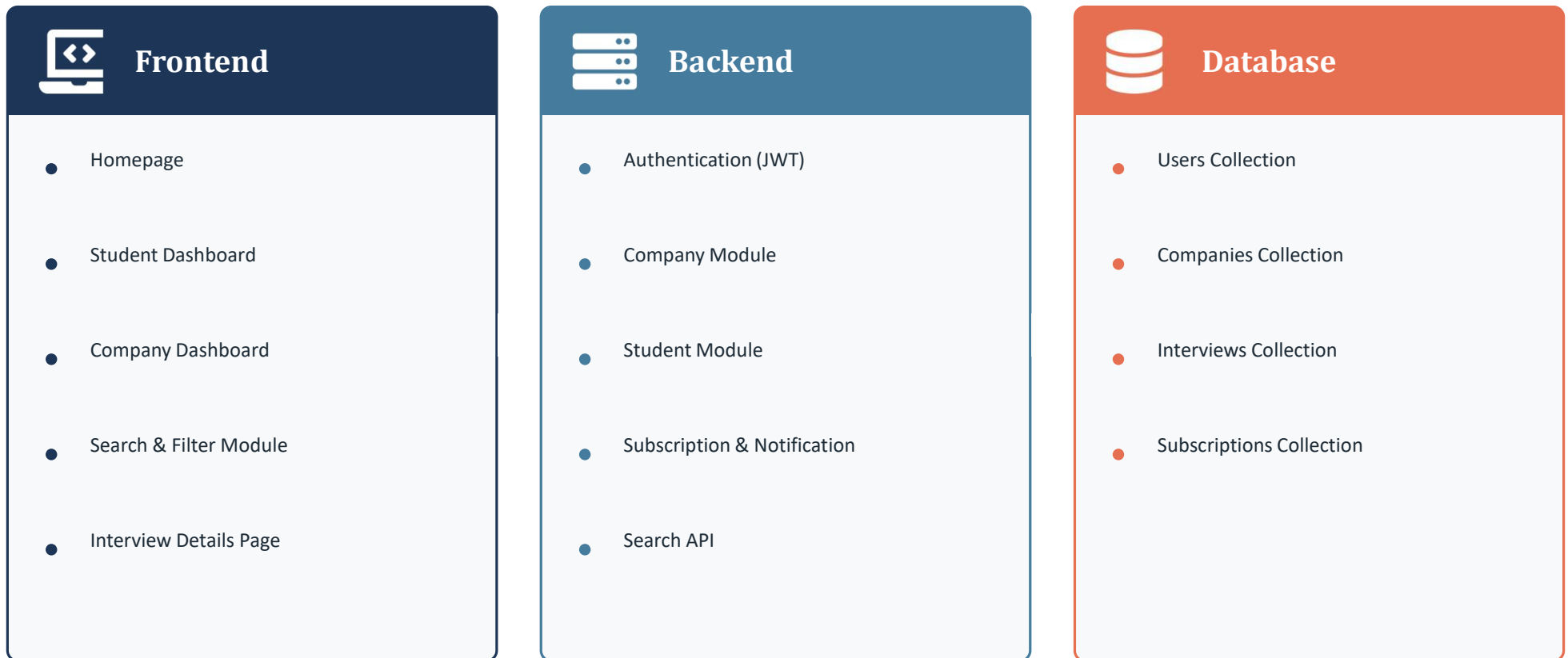
Postman

Vercel

Render / Heroku

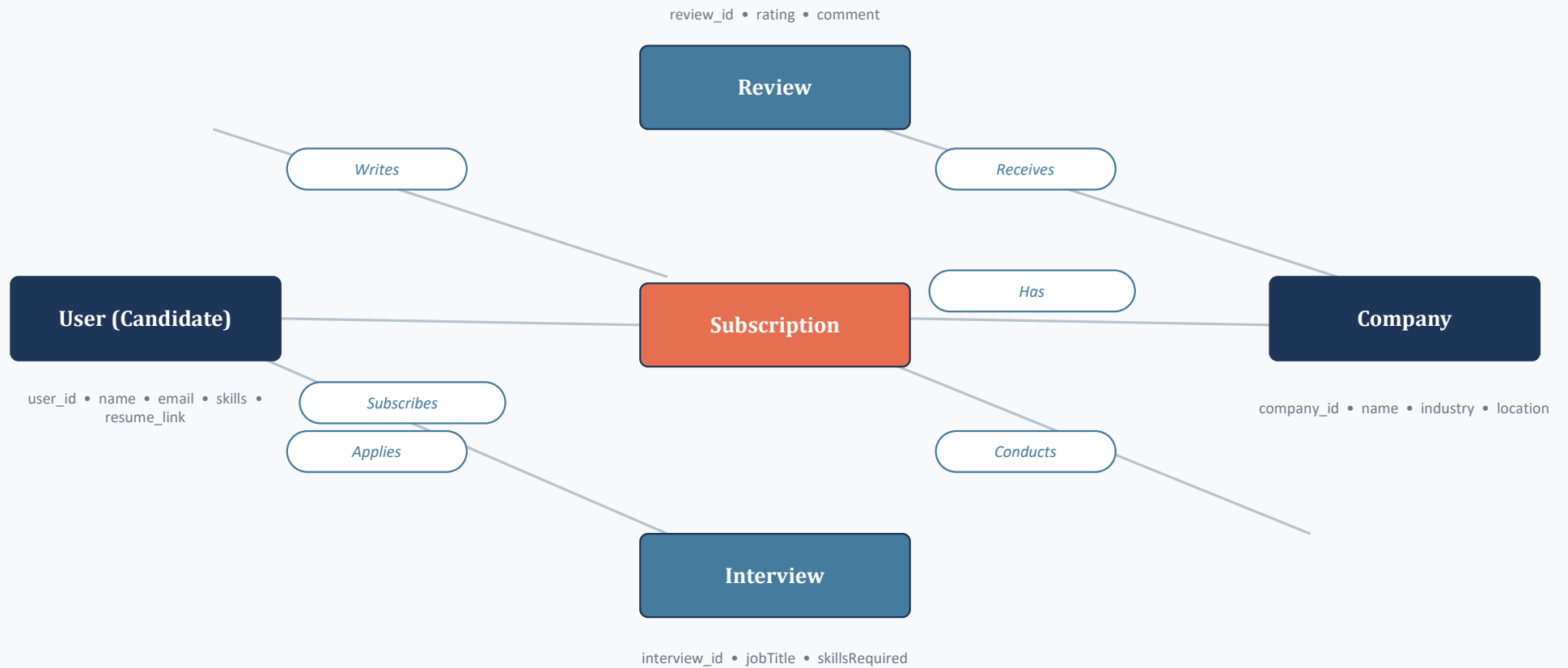
Deployment: Vercel (frontend) • Render / Heroku (backend)

System Architecture



DATABASE DESIGN

Entity Relationship Diagram



DATA FLOW

Data Flow Diagram (Level 0 & 1)

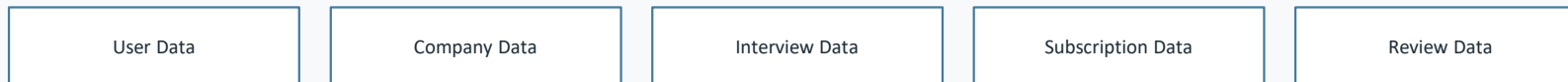
Zero-Level DFD



Level 1 DFD — Major Sub-Processes



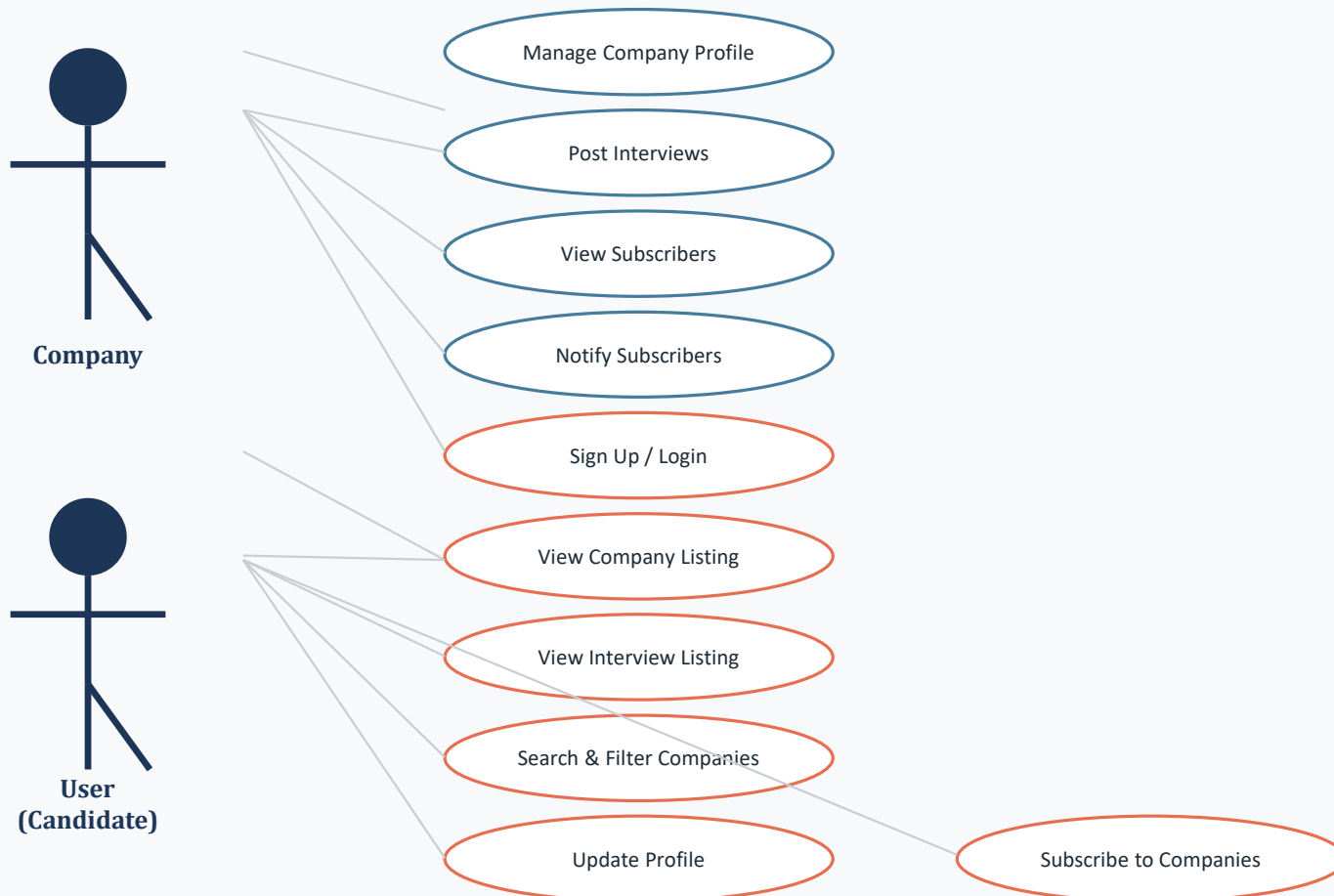
Data Stores



Sub-processes read/write to their respective collections while exchanging query results and confirmations with Users and Companies.

FUNCTIONAL VIEW

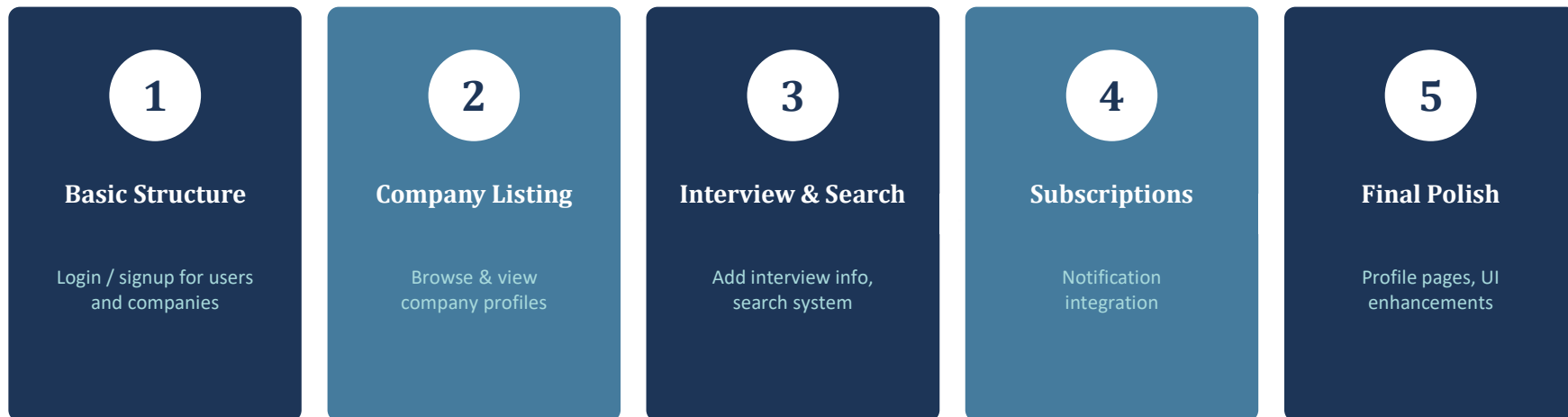
Use Case Diagram



DEVELOPMENT APPROACH

Methodology — Incremental Model

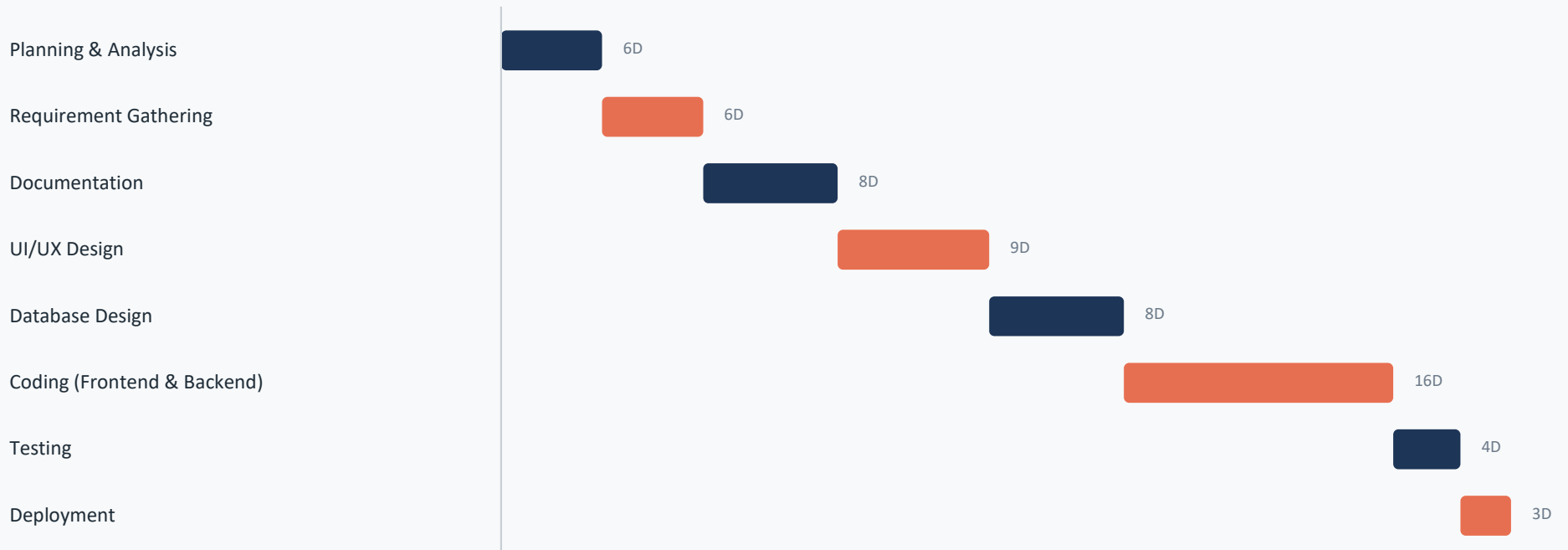
Career Link was built using the Incremental SDLC model — delivering the system in small, testable phases that build on one another.



Each increment is independently tested, then integrated — enabling early testing and continuous improvement.

TIMELINE

Project Planning & Schedule



Total estimated duration: ~60 working days (~2 months)

FUNCTIONAL BREAKDOWN

Key Modules



User Module

(Candidate)

- Authentication — Sign up / Login
- Profile Management — resume, skills
- Company & Interview Listings
- Subscription System
- Review System



Company Module

(Recruiter)

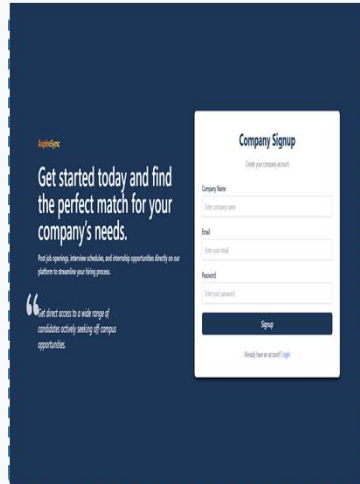
- Company Authentication
- Profile Management — hiring criteria
- Add Interview Details
- Candidate Management
- Notification System

INTERFACE

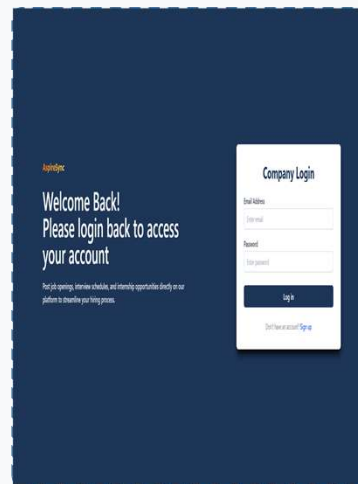
UI Screens (Insert Screenshots)



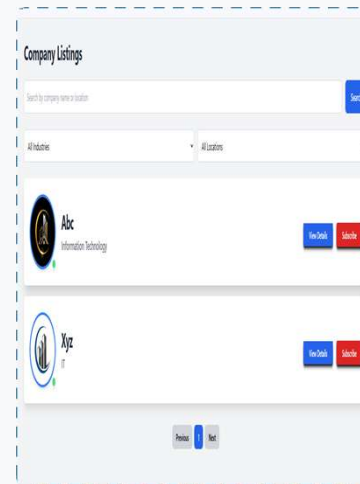
Home Page



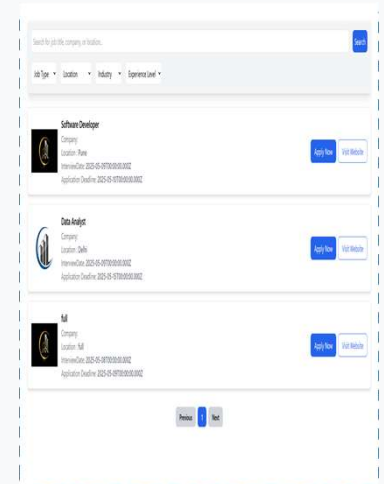
Sign Up Page



Login Page



Company Listing



Interview Listing

Replace each placeholder with the corresponding screenshot from the live application before presenting.

QUALITY ASSURANCE

Testing Strategy



Black Box Testing

Login, subscription & interview features



White Box Testing

JWT auth flow, API routes, backend logic



Unit Testing

Login handler, subscription controller, CRUD



Integration Testing

Frontend-backend communication



Usability Testing

Navigation ease across devices



Performance Testing

Concurrent users & search traffic



Regression Testing

Core features stay intact after updates



Grey Box Testing

Role-based access & secure data flow

COCOMO MODEL

Cost Estimation

Semi-Detached project — estimated using the Basic COCOMO formula: $\text{Effort (PM)} = 3.0 \times (\text{KLOC})^{1.12}$

20

KLOC

Estimated project size

76.4

Person-Months

Theoretical COCOMO effort

320

Hours

Actual effort (2 months, solo)

₹1,60,000

Estimated Cost

At ₹500 / hour developer rate

A single-developer effort over a 2-month period, covering design, coding, testing, and deployment for the complete platform.

CONCLUSION

Expected Outcomes



Fully functional platform connecting students and companies for off-campus hiring



Reduced job-role mismatch through AI-assisted resume matching



Centralized, easy-to-use hiring ecosystem for both sides



Scalable foundation ready for analytics, dashboards & new integrations

Career Link demonstrates a complete MERN-stack solution — from AI-driven matching to a modular, scalable architecture.

Thank You

Questions & Discussion Welcome

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